

FleetGuard Smart Environmental Monitoring System

Component List (Bill of Materials)

Item	Component	Specification	Qty	Purpose
1	ESP32 DevKit V1 (30-pin)	ESP32-WROOM-32	1	Main microcontroller
2	DHT22 Sensor	Temperature & Humidity	1	Environmental sensing
3	OLED Display	0.96in SSD1306 I2C	1	Display readings/status
4	Green LED	5mm	1	Normal indication
5	Yellow LED	5mm	1	Warning indication
6	Red LED	5mm	1	Critical indication
7	220Ω Resistor	1/4W	3	LED current limiting
8	Active Buzzer	3.3V/5V	1	Audible alert
9	MicroSD Module	SPI Interface	1	Local event logging
10	Push Button	Momentary	1	Configuration/Reset
11	5V Input Terminal	2-pin Screw Terminal	1	Power input
12	USB Cable	USB	1	Programming/Power
13	Jumper Headers	2.54 mm	Several	Interconnections
14	Decoupling Capacitors	100nF	4	Power filtering
15	Electrolytic Capacitor	10uF	2	Power stabilization

Estimated Supply: 5V DC input, onboard 3.3V regulation via ESP32 DevKit.

Notes: Suitable for proof-of-concept implementation. GPIO assignments should match the firmware configuration.